OL | Organic Letters

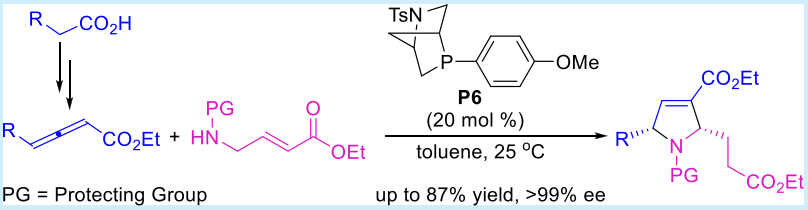
pubs.acs.org/OrgLett Letter

Phosphine-Catalyzed Asymmetric Tandem Isomerization/Annulation of Allyl Amines with Allenoates: Enantioselective Annulation of a Saturated C–N Bond

Leijie Zhou, Xue Zhang, Qijun Wang, Min Liu, Wei Wang, Yongjun Wu, Liezhong Chen,* and Hongchao Guo*

Cite This: *Org. Lett.* 2021, 23, 9173–9178 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: Under catalysis by chiral phosphine, an asymmetric isomerization/annulation cascade reaction of allylamines with allenoates was realized. A wide range of γ-substituted allenoates were tolerated to afford chiral pyrroline derivatives in high yields with excellent enantioselectivities. In the reaction, isomerization of readily available *N*-allylamines to reactive aliphatic imines through a 1,4-proton shift is a key step, which circumvents the isolation of highly unstable alkyl *N*-sulfonylimines.

Downloaded via FUDAN UNIV on January 17, 2022 at 10:26:34 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Enantiopure five-membered N-heterocycles occupy a prominent place in synthetic chemistry, and they are also one of the most common key motifs in numerous bioactive molecules, medicines, and natural products.[1] Thus, development of their synthetic methods is of high value. In the past two decades, phosphine-catalyzed annulation reactions have been proven to be a robust, versatile tool for construction of various cyclic compounds through a cycloaddition mode of zwitterionic intermediates and electrophiles, especially in highly enantioselective ways.[2] As one of the most typical and representative annulation modes in phosphine catalysis, the (3+2) annulation of allenoates and aromatic imines has been investigated thoroughly[3,4] since it was reported first in 1997 by Lu.[5] A large number of elegant methods were established, providing convenient synthetic routes to various pyrroline-cored skeletons (left, Scheme 1a).[3,4] However, due to the isomeric nature and relative instability, aliphatic imines are rarely involved in this enantioselective (3+2) annulation reaction. The only breakthrough in the application of aliphatic imines was made in 2012 by Lu (middle, Scheme 1a).[6] Through utilizing the diphenylphosphinoyl (DPP) group as the protecting group of imines and amino acid-based chiral phosphines as the catalysts, both high yields and enantioselectivities were achieved with a broad range of aryl and alkyl imines. However, the potential of variation at position 5 of pyrroline products had not been evaluated. It is well-known that synthetic chemistry greatly depends on the strategy of preinstallations and sequence couplings and creates the core chiral skeleton, in which preinstallations need to be performed in flexible alkyl substituents. Thus, simultaneously enabling alkyl variations at both positions 2 and 5 is highly valuable for the application of this annulation mode in chemical synthesis (right, Scheme 1a).

N-Allylamines represent a class of significant raw materials and key intermediates in medicinal chemistry and the pharmaceutical industry. Their transformations and application have attracted a great deal of attention.[7,8] To circumvent the challenge in preparing unstable alkyl imines, capturing reactive alkyl imines in situ from isomerization of allyl amines is an attractive strategy, which enables some previously elusive design. For instance, the Terada group reported a relay catalysis for a tandem isomerization/carbon–carbon bond formation sequence of allyl amines in 2008 (Scheme 1b).[9] The reaction depended on the generation of reactive imines from readily available allyl amines in a one-pot reaction via isomerization, which enabled the functionalization at the previously saturated α position. This double bond migration/isomerization strategy inspired us to examine whether this isomerization can be realized in organocatalysis for some challenging transformations. As part of our continuing study of annulation reactions,[10] we herein report chiral phosphine-

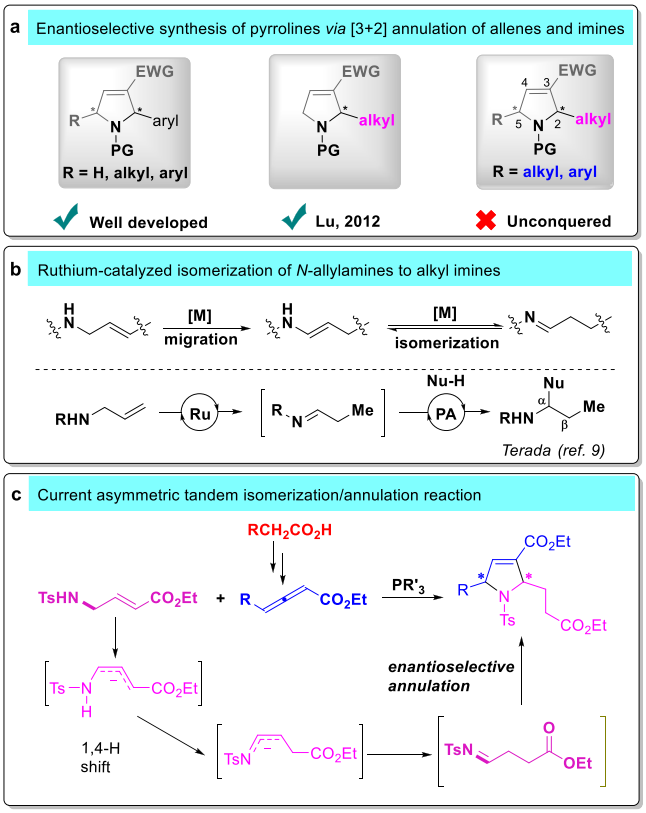
Received: October 14, 2021
Published: November 16, 2021



ACS Publications © 2021 American Chemical Society

https://doi.org/10.1021/acs.orglett.1c03483
Org. Lett. 2021, 23, 9173–9178

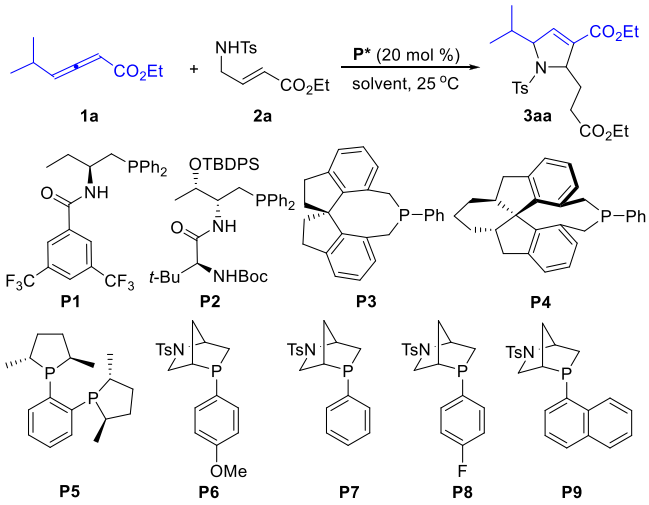
Scheme 1. Pyrrolines and Reactions Involving Imines and Allyl Amines



catalyzed asymmetric tandem isomerization/annulation of allyl amines with allenoates, enabling enantioselective functionalization of a saturated C−N bond (Scheme 1c).[11] As shown in Scheme 1c, generation of reactive alkyl imine and its subsequent capture in situ circumvent the isolation of highly unstable alkyl *N*-sulfonylimines and are key steps.

At the outset of our design, we chose the reaction between γ-isopropyl-substituted allenoate **1a** and γ-sulfonamido-substituted vinyl ester **2a** as the model reaction. As displayed in Table 1, without the use of phosphine as the catalyst, no reaction occurred (entry 1). In the presence of achiral phosphine $PPhMe_2$, the reaction worked to give the anticipated product in 65% yield (entry 2). Various chiral phosphines such as amino acid-based bifunctional chiral phosphines (**P1** and **P2**), SPINOL-based chiral phosphines (**P3** and **P4**), and chiral bisphosphine **P5** were investigated, in which only bifunctional chiral phosphine **P2** showed a relatively low catalytic activity with poor enantioselectivity (entries 3−7). To our delight, when we turned to the commercially available bicyclic chiral phosphine **P6**, the desired product **3aa** was obtained in a moderate yield with excellent enantioselectivity (entry 8). Inspired by this result, we changed the solvent to toluene and reversed the **1a**:**2a** substrate ratio and thus optimized the result to 75% yield and 97% ee (entries 9 and 10). Further adjustment of the substrate ratio caused a minor decrease in catalytic activity (entries 11 and 12). Other analogues (**P7**−**P9**) of chiral phosphine **P6** led to significant decreases in yields and minor decreases in enantioselectivities (entries 13−15, respectively). Catalysts **P3**−**P9** all belong to the dialkyl arylphosphine group; however, there are huge distinctions in the catalytic reactivities. The reason might be their different three-dimensional structure.

Table 1. Optimization of Reaction Conditions[a]



entry	catalyst	**1a**:**2a** ratio	solvent	*t* (h)	yield (%)[b]	ee (%)[c]
1	none	2:1	toluene	72	0	–
2	$PPhMe_2$	2:1	toluene	4	65	–
3	**P1**	2:1	CH_2Cl_2	72	trace	–
4	**P2**	2:1	CH_2Cl_2	72	40	28
5	**P3**	2:1	CH_2Cl_2	72	trace	–
6	**P4**	2:1	CH_2Cl_2	72	trace	–
7	**P5**	2:1	CH_2Cl_2	72	trace	–
8	**P6**	2:1	CH_2Cl_2	24	43	95
9	**P6**	2:1	toluene	24	46	97
10	**P6**	1:2	toluene	24	75	97
11	**P6**	1:1.5	toluene	24	72	97
12	**P6**	1:1.2	toluene	24	69	97
13	**P7**	1:2	toluene	24	49	94
14	**P8**	1:2	toluene	24	47	94
15	**P9**	1:2	toluene	24	58	96

[a]Unless indicated otherwise, reactions of **1a** (0.20 mmol) and **2a** (0.10 mmol) were carried out at 25 °C in the presence of a catalyst (0.02 mmol) in 1 mL of solvent. [b]Isolated yield. The dr is >20:1, determined by ^{1}H NMR analysis. [c]Determined by chiral HPLC analysis.

With the optimized conditions in hand, γ-substituted allenoates, which were easily prepared from various substituted acetic acids (see the Supporting Information), were examined. As displayed in Figure 1, the results indicated that methyl or branched alkyl substituents at the γ position of allenoate seem to have a remarkably positive influence on the reaction. For example, the allenoate having isopropyl, methyl, or *tert*-butyl at the γ position worked well in this reaction with high yield and excellent enantioselectivity (**3aa**−**3ac**). However, allenoates with linear alkyl substituents at the γ position exhibited much less reactivity, while retaining the excellent enantioselectivity (**3ad** and **3ae**). The branched alkyl-substituted allenoates with a stereogenic center smoothly afforded the corresponding dihydropyrrole derivatives (**3af**−**3ai**) in moderate to good yields with excellent enantioselectivities, despite poor diastereoselectivities. Unfortunately, diphenyl- and triphenyl-substituted allenoates had no reactivity in this reaction, due to unfavorable electronic properties (**3aj** and **3ak**). Cyclopentylmethyl-substituted allenoate can also afford the desired product (**3al**) with excellent enantioselectivities, albeit in relatively low yields. In particular, cycloalkyl- and bridged cycloalkyl-substituted allenoates exhibited both high reactivity

https://doi.org/10.1021/acs.orglett.1c03483
Org. Lett. 2021, 23, 9173−9178

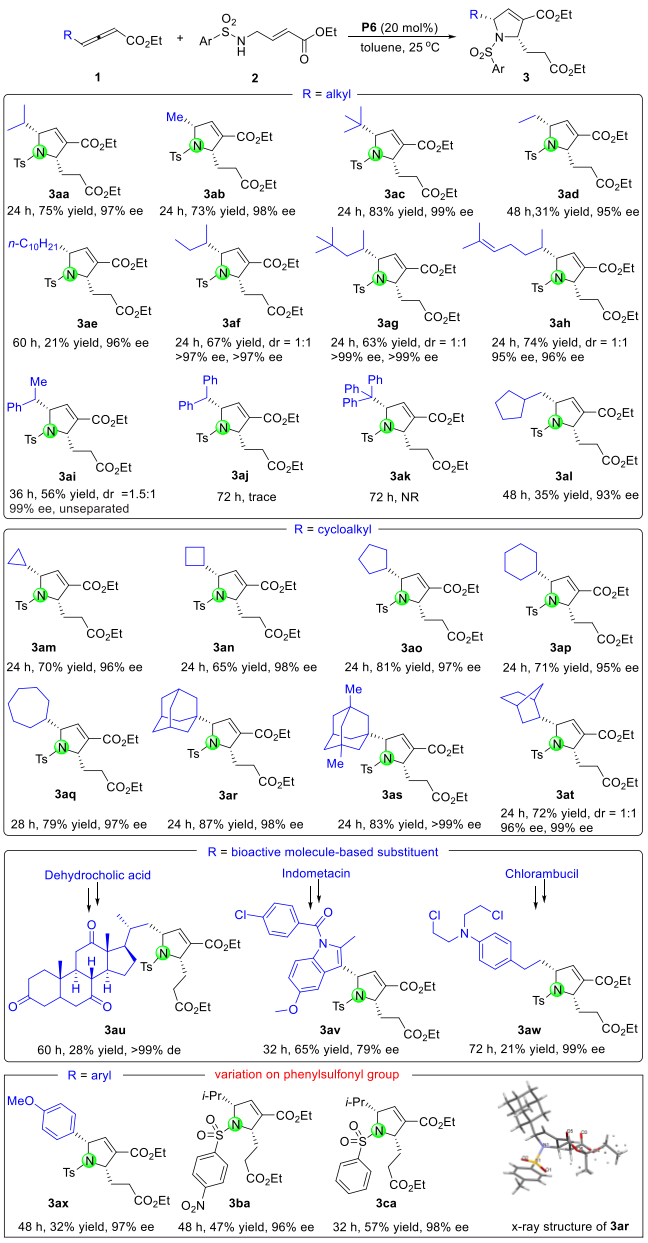
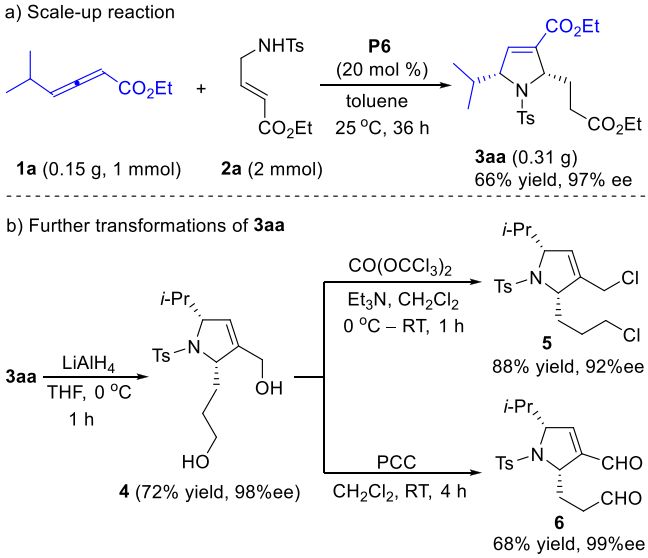


Figure 1. Investigation of the scope of chiral alcohols. Unless indicated otherwise, reactions of **1** (0.10 mmol) and **2** (0.20 mmol) were carried out at 25 °C in the presence of **P6** (0.02 mmol) in 1 mL of toluene. The dr is >20:1, determined by ^{1}H NMR analysis. Yields are isolated yields, and ee values were determined by chiral HPLC analysis.

(65−87% yields) and superior enantioselectivity (95−>99% ee) to provide the corresponding dihydropyrrole products (**3am**−**3at**), which further proved that the branched alkyl substituents at the γ position of allenoates led to higher reactivity. Bioactive molecule-derived allenoates were also tested in this reaction. For example, the common drug dehydrocholic acid-derived allenoate afforded the corresponding product **3au** in >99% de. In contrast, the drug indomethacin-derived allenoate provided product **3av** in 65% yield and 79% ee. The other drug chlorambucil-derived allenoate gave product **3aw** in 99% ee. Unlike cycloalkyl-substituted allenoates, aryl-substituted allenoate showed low reactivity despite the superior enantioselectivity (**3ax**). In addition, variations of the arylsulfonyl group of γ-sulfonamido-substituted vinyl esters **2** may be tolerated, giving the corresponding products **3ba** and **3ca** in satisfactory yields. The absolute configurations of products **3** were confirmed by analogy to the results of single-crystal X-ray diffraction analysis of product **3ar** (detailed in the Supporting Information).

As demonstrated in Scheme 2a, this phosphine-catalyzed (3+2) annulation reaction is scalable, being able to accomplish

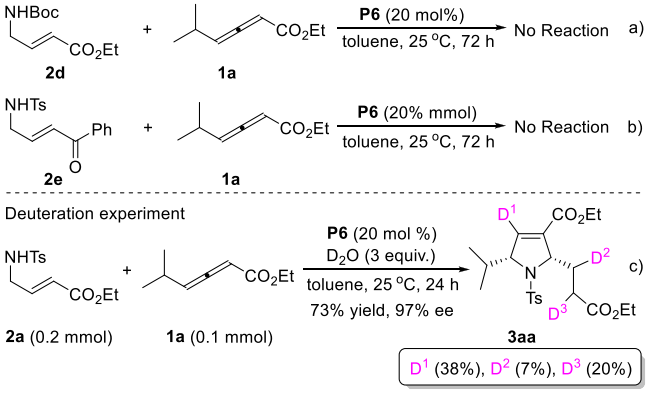
Scheme 2. Scale-up Reaction and Further Transformations of Product 3aa



the synthesis of cycloadduct **3aa** in 65% yield and 97% ee. Scheme 2b reveals that the product may perform diverse transformations, while treatment of product **3aa** with $LiAlH_4$ achieved reduction of the ester groups, affording diol derivative **4** in 72% yield and 98% ee. Further treatment of diol **4** with triphosgene failed to afford an envisaged nine-membered ring but generated dichloride derivative **5** in 88% yield and 92% ee. Direct oxidation of product **4** by PCC produced dialdehyde **6** in 68% yield and 99% ee (Scheme 2b).

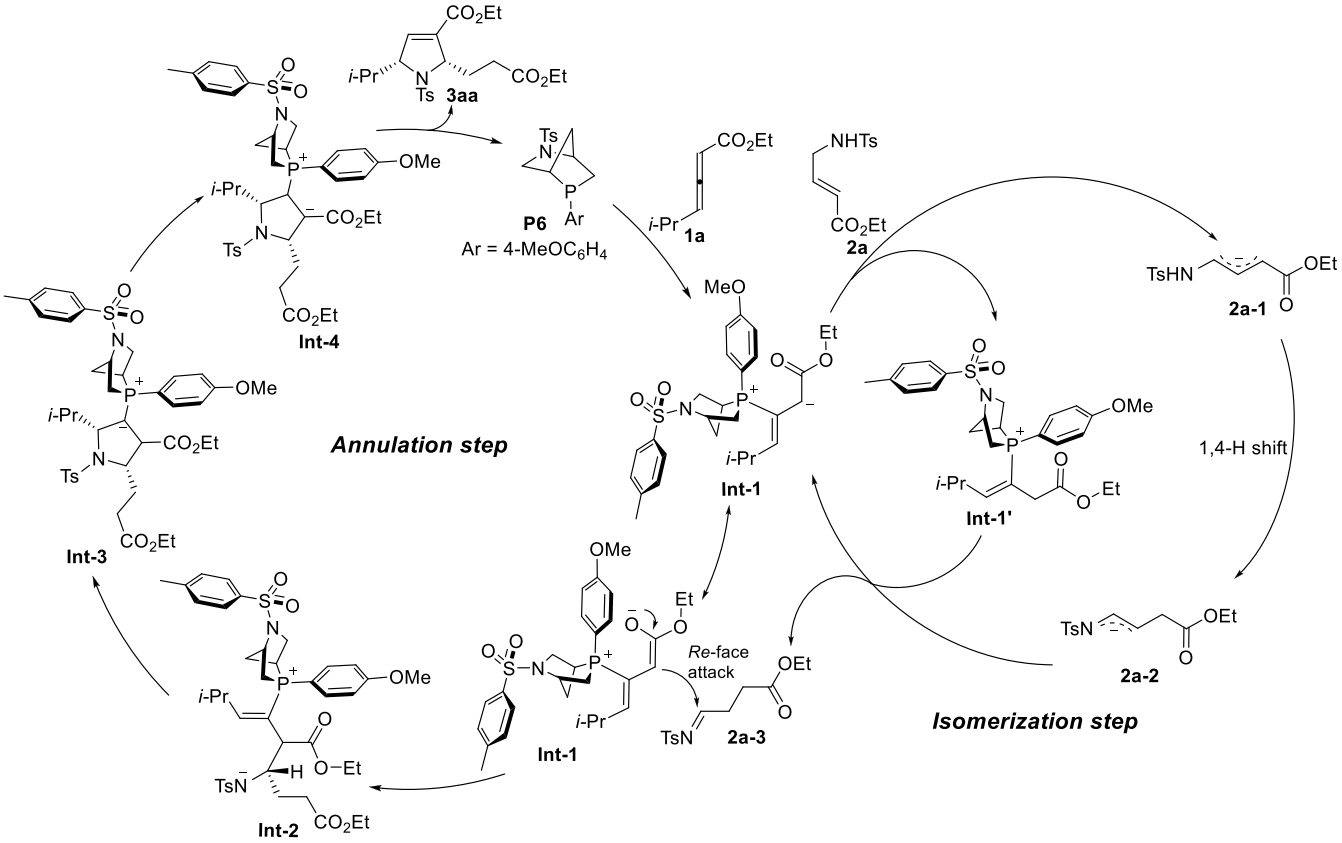
To further explore the reaction mechanism, some control experiments were carried out (Scheme 3). When the sulfonyl protecting group of **1a** was changed to Boc, the reaction did not occur (Scheme 3a). Replacement of the ester group with the benzoyl group led to no reactivity (Scheme 3b). All of the results verified that the unique properties of the sulfonyl group

Scheme 3. Control Experiments



https://doi.org/10.1021/acs.orglett.1c03483
Org. Lett. 2021, 23, 9173−9178

Scheme 4. A Plausible Mechanism



and ester group have a crucial influence on this annulation reaction. A deuteration experiment was also performed to gain further insights into the proton shift (Scheme 3c). Adding 3 equiv of D_2O to the model reaction mixture under optimized reaction conditions led to the successful incorporation of deuterium into the product, which indicated that an isomerization of substrate **2a** definitely occurred.

As shown in Scheme 4, a plausible mechanism for this tandem isomerization/annulation reaction was proposed on the basis of results of control experiments and previous studies.[2−4,11] First, phosphine **P6** attacked allenoate **1a** to form zwitterionic intermediate **Int-1**. This zwitterion acted as a base to deprotonate 4-sulfonamido-2-butenoate **2a** to give intermediates **2a-1** and **Int-1′**. **2a-1** performed a 1,4-proton shift to afford enamine anion **2a-2**, which could obtain a proton from **Int-1′** to give imine **2a-3** with simultaneous regeneration of intermediate **Int-1**. Intermediate **Int-1** worked as a nucleophilic reagent for an attack on the C═N bond of **2a-3** from the *re*-face to generate **Int-2**, which then performed intramolecular addition to furnish annulation, producing intermediate **Int-3**. A subsequent 1,2-proton shift followed by the elimination of the catalyst led to product **3aa**. The regenerated catalyst continued to participate in the catalytic circle. The results of DFT calculation demonstrated the rationality of the proton shift in the catalytic process (see the Supporting Information for details). It is worth noting that only α-addition products were observed via this method. According to DFT calculations performed by Dudding,[12] the regiochemical outcome of the current reaction involved Lewis acid activation, van der Waals interactions, and steric repulsion, and α addition was favored in terms of energy, compared with γ addition.

In summary, we have developed an unprecedented tandem isomerization/annulation sequence of allylamines and γ-substituted allenoates through phosphine catalysis. This reaction enabled the generation of reactive aliphatic imines from readily available *N*-sulfonyl allylamine under phosphine catalysis, thus realizing the challenging asymmetric (3+2) annulation between aliphatic imines and γ-substituted allenoates. A series of biologically important chiral pyrroline derivatives were obtained in high yields and excellent enantioselectivities. A wide scope of γ-substituted allenoates prepared efficiently from various substituted acetic acid compounds were well tolerated under the mild reaction conditions; in particular, those bioactive molecule-based allenoates prepared from specific drugs containing acetic acid units also afforded good results. Further transformations of the products showed their great potential in construction of various bioactive enantiopure compounds. In addition, the experimental results and DFT calculation demonstrated that isomerization of allylamines to imines played a key role in the reaction and the 1,4-proton shift was favorable. Further studies of the usability of this reaction in the synthesis of heterocyclic compounds for pesticide research are currently underway in our laboratory and will be reported in due course.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c03483.

Experimental procedure, characterization data, HPLC analysis data, NMR spectra, and X-ray crystallographic data (PDF)

https://doi.org/10.1021/acs.orglett.1c03483
Org. Lett. 2021, 23, 9173−9178

Accession Codes

CCDC 1988551 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

AUTHOR INFORMATION

Corresponding Authors

Hongchao Guo − *Department of Chemistry and Innovation Center of Pesticide Research, China Agricultural University, Beijing 100193, P. R. China;* orcid.org/0000-0002-7356-4283; Email: hchguo@cau.edu.cn

Liezhong Chen − *State Key Laboratory for Managing Biotic and Chemical Threats to the Quality and Safety of Agro-products, Zhejiang Academy of Agricultural Sciences, Hangzhou 310021, China*; Email: zwsclz@163.com

Authors

Leijie Zhou − *Department of Chemistry and Innovation Center of Pesticide Research, China Agricultural University, Beijing 100193, P. R. China*

Xue Zhang − *State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Shanghai 200032, China*

Qijun Wang − *Department of Chemistry and Innovation Center of Pesticide Research, China Agricultural University, Beijing 100193, P. R. China*

Min Liu − *Department of Chemistry and Innovation Center of Pesticide Research, China Agricultural University, Beijing 100193, P. R. China*

Wei Wang − *College of Public Health, Zhengzhou University, Zhengzhou 450001, P. R. China*

Yongjun Wu − *College of Public Health, Zhengzhou University, Zhengzhou 450001, P. R. China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.1c03483

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was funded by the National Natural Science Foundation of China (21871293 and 22071264) and the National Key R&D Program of China (2018YFD0200100).

REFERENCES

(1) (a) Coldham, I.; Hufton, R. Intramolecular dipolar cycloaddition reactions of azomethine ylides. *Chem. Rev.* **2005**, *105*, 2765−2810. (b) Pandey, G.; Banerjee, P.; Gadre, S. R. Construction of enantiopure pyrrolidine ring system via asymmetric [3 + 2]-cycloaddition of azomethine ylides. *Chem. Rev.* **2006**, *106*, 4484−4517.

(2) For selected reviews of phophine catalysis, see: (a) Wang, Z.; Xu, X.; Kwon, O. Phosphine catalysis of allenes with electrophiles. *Chem. Soc. Rev.* **2014**, *43*, 2927−2940. (b) Xiao, Y.; Sun, Z.; Guo, H.; Kwon, O. Chiral phosphines in nucleophilic organocatalysis. *Beilstein J. Org. Chem.* **2014**, *10*, 2089−2121. (c) Li, W.; Zhang, J. Recent developments in the synthesis and utilization of chiral β-aminophosphine derivatives as catalysts or ligands. *Chem. Soc. Rev.* **2016**, *45*, 1657−1677. (d) Xiao, Y.; Guo, H.; Kwon, O. Nucleophilic chiral phosphines: Powerful and versatile catalysts for asymmetric annulations. *Aldrichimica Acta* **2016**, *49*, 3. (e) Wei, Y.; Shi, M. Lu's [3 + 2] cycloaddition of allenes with electrophiles: discovery, development and synthetic application. *Org. Chem. Front.* **2017**, *4*, 1876−1890. (f) Ni, H.; Chan, W. L.; Lu, Y. Phosphine-catalyzed asymmetric organic reactions. *Chem. Rev.* **2018**, *118*, 9344−9411. (g) Guo, H.; Fan, Y. C.; Sun, Z.; Wu, Y.; Kwon, O. Phosphine organocatalysis. *Chem. Rev.* **2018**, *118*, 10049−10293. (h) Li, E. Q.; Huang, Y. Recent advances in phosphine catalysis involving γ-substituted allenoates. *Chem. Commun.* **2020**, *56*, 680−694. (i) Huang, Y.; Liao, J.; Wang, W.; Liu, H.; Guo, H. Synthesis of heterocyclic compounds through nucleophilic phosphine catalysis. *Chem. Commun.* **2020**, *56*, 15235−15281.

(3) For selected examples of [3+2] annulation of allenoates and aromatic imines, see: (a) Zhu, X. F.; Henry, C. E.; Kwon, O. A highly diastereoselective synthesis of 3-carbethoxy-2, 5-disubstituted-3-pyrrolines by phosphine catalysis. *Tetrahedron* **2005**, *61*, 6276−6282. (b) Zhao, G. L.; Shi, M. Aza-Baylis-Hillman Reactions of N-Tosylated Aldimines with Activated Allenes and Alkynes in the Presence of Various Lewis Base Promoters. *J. Org. Chem.* **2005**, *70*, 9975−9984. (c) Tang, X.; Zhang, B.; He, Z.; Gao, R.; He, Z. 1, 3, 5-Triaza-7-phosphaadamantane (PTA): A Practical and Versatile Nucleophilic Phosphine Organocatalyst. *Adv. Synth. Catal.* **2007**, *349*, 2007−2017. (d) Zheng, S.; Lu, X. A phosphine-catalyzed [3+ 2] annulation reaction of modified allylic compounds and N-tosylimines. *Org. Lett.* **2008**, *10*, 4481−4484. (e) Sampath, M.; Lee, P. Y. B.; Loh, T. P. Phosphine-catalyzed one-pot isomerization of 3-alkynoates and [2 + 3] cycloaddition with imines: formal synthesis of Securinega alkaloid (±)-allosecurinine. *Chem. Sci.* **2011**, *2*, 1988−1991. (f) Yu, H.; Zhang, L.; Yang, Z.; Li, Z.; Zhao, Y.; Xiao, Y.; Guo, H. Phosphine-catalyzed [3 + 2] cycloaddition of sulfamate-derived cyclic imines with allenoate: Synthesis of sulfamate-fused dihydropyrroles. *J. Org. Chem.* **2013**, *78*, 8427−8436. (g) Zhang, L.; Yu, H.; Yang, Z.; Liu, H.; Li, Z.; Guo, J.; Xiao, Y.; Guo, H. Phosphine-catalyzed [3 + 2] cycloaddition of Morita−Baylis−Hillman carbonates with sulfamate-derived cyclic imines. *Org. Biomol. Chem.* **2013**, *11*, 8235−8240.

(4) For selected examples of asymmetric [3+2] annulation of allenoates and aromatic imines, see: (a) Jean, L.; Marinetti, A. Phosphine-catalyzed enantioselective [3 + 2] annulations of 2,3-butadienoates with imines. *Tetrahedron Lett.* **2006**, *47*, 2141−2145. (b) Fang, Y. Q.; Jacobsen, E. N. Cooperative, highly enantioselective phosphinothiourea catalysis of imine-allene [3 + 2] cycloadditions. *J. Am. Chem. Soc.* **2008**, *130*, 5660−5661. (c) Andrews, I. P.; Kwon, O. Enantioselective total synthesis of (+)-ibophyllidine via an asymmetric phosphine-catalyzed [3 + 2] annulation. *Chem. Sci.* **2012**, *3*, 2510−2514. (d) Henry, C. E.; Xu, Q.; Fan, Y. C.; Martin, T. J.; Belding, L.; Dudding, T.; Kwon, O. Hydroxyproline-derived pseudoenantiomeric [2.2.1] bicyclic phosphines: asymmetric synthesis of (+)-and (−)-pyrrolines. *J. Am. Chem. Soc.* **2014**, *136*, 11890−11893. (e) Cai, L.; Zhang, K.; Kwon, O. Catalytic asymmetric total synthesis of (−)-actinophyllic acid. *J. Am. Chem. Soc.* **2016**, *138*, 3298−3301. (f) Sankar, M. G.; Garcia-Castro, M.; Golz, C.; Strohmann, C.; Kumar, K. Engaging Allene-Derived Zwitterions in an Unprecedented Mode of Asymmetric [3 + 2]-Annulation Reaction. *Angew. Chem., Int. Ed.* **2016**, *55*, 9709−9713. (g) Han, X.; Chan, W. L.; Yao, W.; Wang, Y.; Lu, Y. Phosphine-mediated Highly Enantioselective Spirocyclization with Ketimines as Substrates. *Angew. Chem., Int. Ed.* **2016**, *55*, 6492−6496.

(5) (a) Xu, Z.; Lu, X. Phosphine-catalyzed [3 + 2] cycloaddition reaction of methyl 2, 3-butadienoate and N-tosylimines. A novel approach to nitrogen heterocycles. *Tetrahedron Lett.* **1997**, *38*, 3461−3464. (b) Xu, Z.; Lu, X. A novel [3 + 2] cycloaddition approach to nitrogen heterocycles via phosphine-catalyzed reactions of 2, 3-butadienoates or 2-butynoates and dimethyl acetylenedicarboxylate with imines: A convenient synthesis of pentabromopseudilin. *J. Org. Chem.* **1998**, *63*, 5031−5041.

(6) Han, X.; Zhong, F.; Wang, Y.; Lu, Y. Versatile enantioselective [3 + 2] cyclization between imines and allenoates catalyzed by dipeptide-based phosphines. *Angew. Chem., Int. Ed.* **2012**, *51*, 767−770.

(7) For a review of the isomerization of allyl amines, see: Krompiec, S.; Krompiec, M.; Penczek, R.; Ignasiak, H. Double bond migration in N-allylic systems catalyzed by transition metal complexes. *Coord. Chem. Rev.* **2008**, *252*, 1819−1841.
(8) For selected examples of the isomerization of allyl amines, see: (a) Novak, B. M.; Cafmeyer, J. T. Meta-stable enamines: Synthesis of simple enamines via catalytic isomerization of allylic amine substrates and their polymerization behavior. *J. Am. Chem. Soc.* **2001**, *123*, 11083−11084. (b) Arisawa, M.; Terada, Y.; Nakagawa, M.; Nishida, A. Selective Isomerization of a Terminal Olefin Catalyzed by a Ruthenium Complex: The Synthesis of Indoles through Ring-Closing Metathesis. *Angew. Chem., Int. Ed.* **2002**, *41*, 4732−4734. (c) Hong, S. H.; Sanders, D. P.; Lee, C. W.; Grubbs, R. H. Prevention of undesirable isomerization during olefin metathesis. *J. Am. Chem. Soc.* **2005**, *127*, 17160−17161.
(9) Sorimachi, K.; Terada, M. Relay Catalysis by a Metal-Complex/Brønsted Acid Binary System in a Tandem Isomerization/Carbon-Carbon Bond Forming Sequence. *J. Am. Chem. Soc.* **2008**, *130*, 14452−14453.
(10) (a) Na, R.; Jing, C.; Xu, Q.; Jiang, H.; Wu, X.; Shi, J.; Zhong, J.; Wang, M.; Benitez, D.; Tkatchouk, E.; Goddard, W. A., III.; Guo, H.; Kwon, O. Phosphine-catalyzed annulations of azomethine imines: allene-dependent [3 + 2], [3 + 3], [4 + 3], and [3 + 2+3] pathways. *J. Am. Chem. Soc.* **2011**, *133*, 13337−13348. (b) Jing, C.; Na, R.; Wang, B.; Liu, H.; Zhang, L.; Liu, J.; Wang, M.; Zhong, J.; Kwon, O.; Guo, H. Phosphine-Catalyzed [3 + 2] and [4 + 3]Annulation Reactions of C,N-Cyclic Azomethine Imines with Allenoates. *Adv. Synth. Catal.* **2012**, *354*, 1023−1034. (c) Guo, H.; Liu, H.; Zhu, F.-L.; Na, R.; Jiang, H.; Wu, Y.; Zhang, L.; Li, Z.; Yu, H.; Wang, B.; Xiao, Y.; Hu, X.-P.; Wang, M. Enantioselective Copper-Catalyzed [3 + 3] Cycloaddition of Azomethine Ylides with Azomethine Imines. *Angew. Chem., Int. Ed.* **2013**, *52*, 12641−12645. (d) Liu, H.; Wu, Y.; Zhao, Y.; Li, Z.; Zhang, L.; Yang, W.; Jiang, H.; Jing, C.; Yu, H.; Wang, B.; Xiao, Y.; Guo, H. Metal-Catalyzed [6 + 3] Cycloaddition of Tropone with Azomethine Ylides: A Practical Access to Piperidine-Fused Bicyclic Heterocycles. *J. Am. Chem. Soc.* **2014**, *136*, 2625−2629. (e) Yang, W.; Sun, W.; Zhang, C.; Wang, Q.; Guo, Z.; Mao, B.; Liao, J.; Guo, H. Lewis-Base-Catalyzed Asymmetric [3 + 3] Annulation Reaction of Morita-Baylis-Hillman Carbonates: Enantioselective Synthesis of Spirocyclohexenes. *ACS Catal.* **2017**, *7*, 3142−3146.
(11) During the submission of our manuscript, a similar non-asymmetric reaction was reported: Maddigan-Wyatt, J. T.; Blyth, M. T.; Ametovski, J.; Coote, M. L.; Hooper, J. F.; Lupton, D. W. Redox Isomerization/(3 + 2) Allenoate Annulation by Auto-Tandem Phosphine Catalysis. *Chem. - Eur. J.* **2021**, DOI: 10.1002/chem.202103224.
(12) Dudding, T.; Kwon, O.; Mercier, E. Theoretical rationale for regioselection in phosphine-catalyzed allenoate additions to acrylates, imines, and aldehydes. *Org. Lett.* **2006**, *8*, 3643−3646.

https://doi.org/10.1021/acs.orglett.1c03483
Org. Lett. 2021, 23, 9173−9178